

UNIVERSITY OF DHAKA
Department of Computer Science and Engineering

CSE 4101 — Artificial Intelligence
Lab Report 04 — Population-Based Approach and Swarm Intelligence

Shaolin Swarm Soccer

Emergent Team Play from Boids, Ant Colony Optimization
and Particle Swarm Optimization:

Ten Weak Agents versus a Stronger Defense

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Abstract

This report asks a single question: *can coordination structure alone beat individual quality?* We build a deterministic, dependency-free soccer micro-world in which every attacker is deliberately worse than every defender — slower, myopic, and increasingly inaccurate with kick distance — and we run the *same ten bodies* under two controllers. The *selfish* baseline is individual optimization: chase the ball, dribble at goal, shoot when in range. The *swarm* controller changes nothing about the players and everything about their coordination, combining three classic population-based mechanisms from the lecture: Boids flocking rules for off-ball shape (Reynolds, 1987), Ant Colony Optimization stigmergy for pass-lane selection (Dorigo, 1992), and Particle Swarm Optimization memory for positioning targets (Kennedy & Eberhart, 1995). Aggregated over $3 \text{ seeds} \times 100 \text{ episodes}$, the swarm scores **17/300** goals against **1/300** for the selfish baseline — a $17\times$ improvement with identical players — while completing **95.3%** of its passes. Parameter sweeps show the swarm’s advantage is robust: it widens as kick noise grows, survives strong defenses, and appears only above a critical team size, exactly as swarm-intelligence theory predicts. The system is reproducible end-to-end: a seeded RNG makes every claim a runnable assertion in a Node test suite.

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1 Introduction

Lecture S09 closes with a provocative idea: *simple individuals following simple rules can produce intelligent group-level behaviour*, with no central controller — ants find shortest paths, birds flock, fish school. The claim is easy to state and easy to believe in the abstract. This lab makes it *falsifiable*. We construct an adversarial environment in which the claim must either show up in a scoreboard or fail publicly.

The vehicle is a soccer micro-world inspired by the film *Shaolin Soccer*: a team of individually hopeless players who win only when they play as one organism. Our attackers are handicapped on every axis that matters — speed, accuracy, vision — against an organized, faster defense with a goalkeeper. We then run *exactly the same bodies* under two controllers and count goals:

- **Selfish baseline** — each player greedily optimizes alone: chase the ball, dribble at goal, shoot the moment the goal is “in range.” This is individual optimization, the natural straw-man of the lecture.
- **Swarm controller** — no attribute of any player changes. Instead, three population-based mechanisms from the lecture coordinate them: *Boids* decides where players stand when they do not have the ball, *ACO stigmergy* decides where the ball goes next, and *PSO* gives the team a spatial memory of where good things happened.

Because the two conditions share every physical parameter and every random seed, any difference in outcome is attributable to coordination structure alone. This is the controlled experiment the lecture’s slogan deserves.

Contributions. (i) A formal problem framing of “weak individuals vs. strong defense” with explicit asymmetry parameters (§3); (ii) a single controller that integrates Boids, ACO and PSO, each mapped to a distinct sub-decision of the team (§5); (iii) a deterministic, zero-dependency implementation with an assert-based test suite in which the headline scientific claim is itself a test (§6); (iv) parameter sweeps over kick noise, evaporation, team size and defender skill that characterize *when* and *why* the collective wins (§7); and (v) a set of negative results — things that should have worked and did not — that connect directly to the exploration/exploitation and diversity lessons of the lecture (§8).

2 Background: The Population-Based Approach (Lecture S09)

The lecture develops population-based search in three stages: Genetic Algorithms, Swarm Intelligence in general, and the two canonical swarm algorithms ACO and PSO. We summarize each and note precisely which conceptual tool this lab borrows from it.

2.1 Genetic Algorithms: vocabulary of population search

A GA attacks an optimization problem with a *population* of candidate solutions rather than a single one. The lecture’s core vocabulary:

- **Representation.** Every candidate is a *chromosome*: a binary string for feature selection, a permutation for routing, a value vector for hyperparameters. A good representation must express valid solutions and let the operators act meaningfully.
- **Fitness.** A fitness function scores each individual and *guides the entire evolutionary process*; it must reflect the real goal and must not mislead the search.

- **Selection.** Better solutions should reproduce more, but weaker ones must not be ignored entirely. The lecture lists *roulette-wheel* selection (probability proportional to fitness), tournament selection, and rank-based selection. Selection pressure controls how greedy the search is.
- **Crossover and mutation.** Crossover *exploits* good existing material; mutation *explores* new regions and prevents stagnation. “Balance is key.”
- **Diversity preservation.** Premature convergence — a population that becomes too uniform and stops improving — is the characteristic GA failure. Remedies include mutation, controlled elitism, moderated selection pressure, and immigration.

This lab is not a GA — nothing here breeds — but it inherits three GA ideas wholesale: *roulette-wheel selection* is exactly how our ball-holder picks a pass lane (§5.2), the *exploration/exploitation balance* turns out to be the difference between an offense that penetrates and one that circulates sterily (§8), and *diversity preservation* appears as a caste split that stops the team from collapsing onto a single consensus point (§5.3).

2.2 Swarm intelligence

Swarm Intelligence (SI) is the branch of AI inspired by collective behaviour of simple agents: ants finding shortest paths, birds flocking, fish schooling, bees foraging. The key idea, quoted from the slides: *intelligence emerges from interaction among many agents, with no single central controller*. SI methods are population-based, flexible, simple, robust, and explore many solutions in parallel — which is why they suit search-intensive problems such as routing, path planning, feature selection and resource allocation.

Our problem is precisely such a search problem in disguise: at every moment the team must find a *route for the ball* through a hostile, moving obstacle field, using only local computations by slow, myopic agents.

2.3 Ant Colony Optimization

ACO is modelled on pheromone trails. Ants deposit pheromone on paths; others preferentially follow stronger trails; shorter paths are completed faster, get reinforced more often, and come to dominate — collective discovery of the optimum through *local* reinforcement. The lecture’s summary loop: initialize pheromone; let many ants construct solutions; evaluate; deposit on good solutions; *evaporate*; repeat. Two concepts carry the whole method: **pheromone** is learned experience, and **evaporation** prevents lock-in to early solutions and maintains exploration.

In this lab the “path” is a passing corridor across the pitch, the “ants” are completed passes, and the shared environment through which the colony communicates — the *stigmergy* medium — is a 25×16 pheromone grid over the field (§5.2).

2.4 Particle Swarm Optimization

PSO is modelled on flocks and schools searching for food. Each candidate solution is a *particle* with a position and velocity in the search space. Every particle learns from two sources of experience: its **personal best** (the best solution it has itself found) and the **global best** (the best found by anyone in the swarm). Velocities are nudged toward both attractors; the swarm converges on promising regions while individual memories keep it spread out.

In this lab each player’s *off-ball positioning target* is the particle. Its personal best is the most dangerous spot where that player ever received a pass; the global best is the team-wide best reception, updated most strongly when a goal is scored (§5.3).

3 Problem Formulation

3.1 The world

The pitch is a rectangle $[0, W] \times [0, H]$ with $W = 100$, $H = 64$ (abstract units). The attacked goal is the segment $x = W$, $y \in [24, 40]$, centre $y_c = 32$. Time advances in discrete *ticks*. An *episode* starts with the ball at midfield (given to the attacker nearest $(50, 32)$) and ends in exactly one of: **goal**, **turnover** (tackle, interception, defender wins loose ball, ball out of bounds), or **timeout** (1200 ticks). The pitch geometry, castes and the pheromone grid are shown in Figure 1.

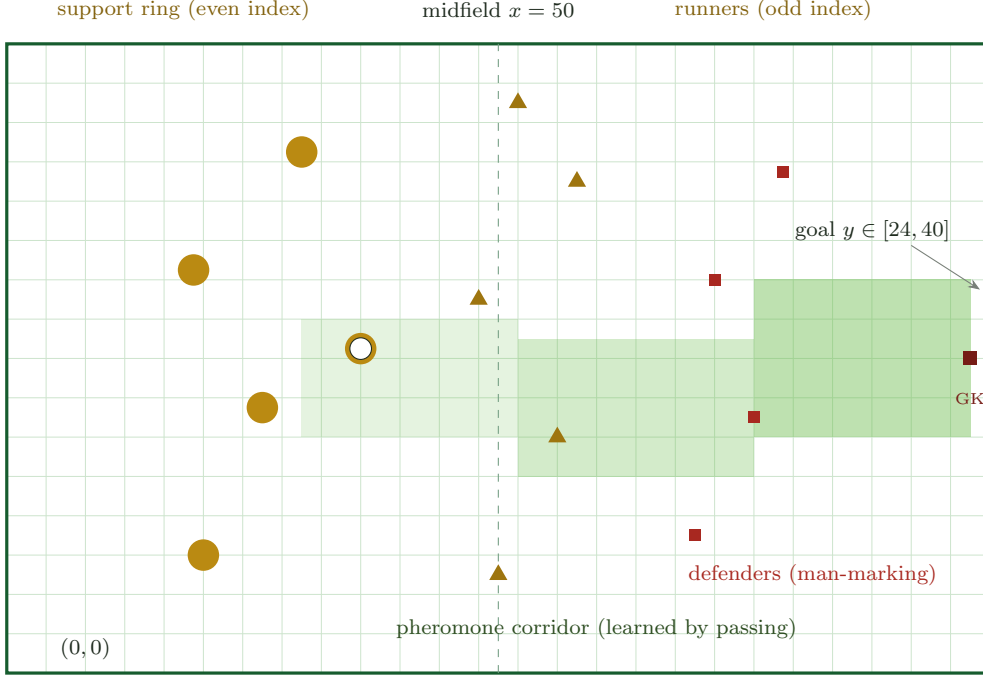


Figure 1: The micro-world: a 100×64 pitch discretized by a 25×16 pheromone grid (cells of 4×4 units). Gold circles are support players, gold triangles are deep runners (the caste split of §5.3), red squares are defenders, and the keeper guards the goal mouth. The green wash sketches a pheromone corridor of the kind visible in the dashboard heatmap: completed passes deposit trail, goals reinforce it, evaporation fades it.

3.2 The asymmetry: every attacker worse than every defender

The experiment is only meaningful if the individuals are genuinely inferior, so the handicaps are explicit and severe (Table 1).

Table 1: Designed asymmetry between attackers and defenders.

Attribute	Shaolin attacker	Defender	Ratio
Top speed	0.5 u/tick (0.38 dribbling)	0.68 u/tick	0.74×
Kick accuracy	$\sigma \propto \text{distance}$, Eq. (1)	—	—
Vision	myopic, radius 15	full field	—
Organization	none built-in	marking + presser + keeper	—
Numbers	10	4 + keeper	2×

Numerical superiority is the attackers' only asset — and exploiting it requires coordination.

The single most important physical law of the world is that **aim error grows linearly with kick distance**. A kick toward a point at distance d has its direction perturbed by a Gaussian

angular error

$$\theta \sim \mathcal{N}(0, \sigma(d)^2), \quad \sigma(d) = 0.011 \cdot d \cdot \eta, \quad (1)$$

where η is the *kick-noise* dial (default 1). Since the lateral miss at the target scales like $d\sigma(d) \propto d^2$, a 60-unit shot is roughly $25\times$ less precise at its target than a 12-unit pass. This one fact creates the ecological niche for cooperation: a relay of short, accurate passes can do what no long heroic ball can (Figure 2).

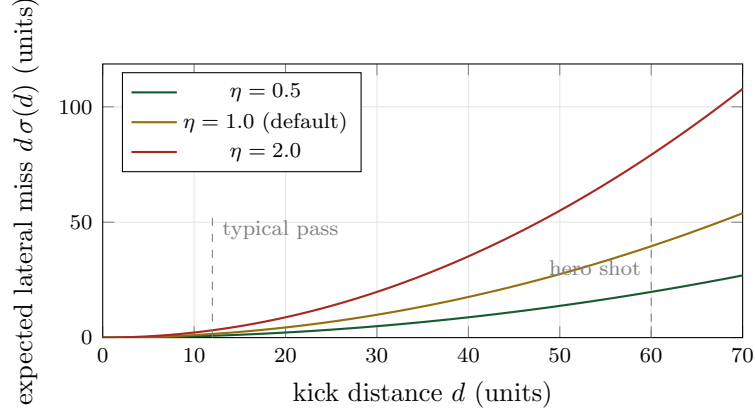


Figure 2: The physics that rewards cooperation: expected lateral miss grows *quadratically* with kick distance (σ linear in d , miss $\approx d\sigma$). Five 12-unit passes stay accurate where one 60-unit punt sprays wide.

3.3 Parameters

Table 2: Complete simulation parameters (defaults; all exposed as UI sliders or constructor options).

Symbol / name	Default	Meaning
$W \times H$	100×64	pitch dimensions
n (nPlayers)	10	attackers
nDefenders	4 (+keeper)	defenders
v_a / v_{drib}	0.5 / 0.38	attacker run / dribble speed (u/tick)
v_d	$0.68 \cdot k$	defender speed; $k = \text{defSkill}$
v_{pass} / v_{shot}	1.7 / 2.6	ball speed for passes / shots
R_{kick}	30	maximum pass range
g (shootGate)	1.15	shoot radius = $g \cdot R_{kick}$
η (kickNoise)	1.0	aim-noise multiplier in Eq. (1)
r_{vis} (vision)	15	attacker vision radius
ρ (evap)	0.0015	pheromone evaporation per tick
grid	25×16	pheromone cells (4×4 units each)
τ_{max}	3	pheromone cap per cell
interceptP	0.22	per-tick interception probability in reach
keeperReach	$2.5 \cdot k$	save radius along the goal mouth
T_{max}	1200	episode timeout (ticks)
R_{react}	$\max(3, \text{round}(14/k))$	defender re-marking period (ticks)
seed	42 (tests: 3, 4, 5)	mulberry32 RNG seed

3.4 Metrics and hypothesis

Per condition we record: **goals**, **shots**, **shots on target**, **pass completion rate** = $\frac{\text{completed}}{\text{attempted}}$, saves, and possession ticks. Bookkeeping is honest by construction: a kicker can never receive his

own pass in flight, and recovering one’s own loose ball counts as a dribble, not a completed pass — without these rules the pass statistics would flatter the swarm.

Hypothesis 1. *With identical bodies and identical physics, the swarm controller scores at least $5\times$ more goals than the selfish controller over 300 episodes ($3 \text{ seeds} \times 100$), and at least 15 in absolute terms.*

Hypothesis 1 is not merely stated — it is literally `assert #4` of the test suite (§6).

4 System Architecture

The system is deliberately minimal: one headless simulation module, one single-file dashboard, one Node test script, zero dependencies. Figure 3 shows the module architecture; Figure 4 shows the per-tick pipeline.

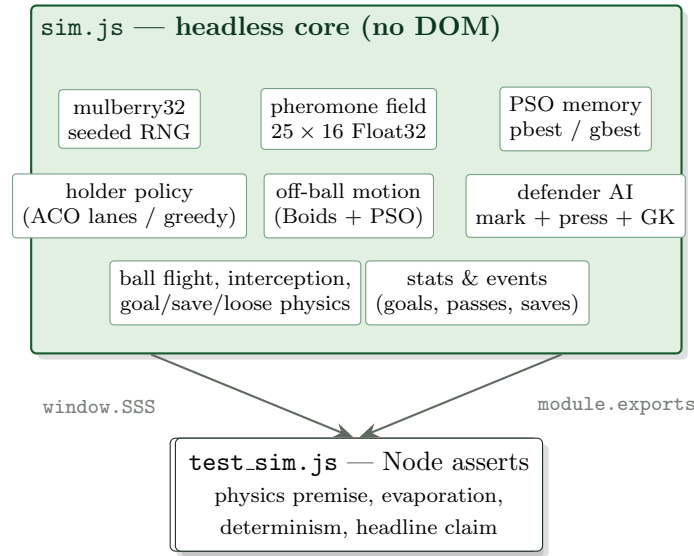


Figure 3: Module architecture. The simulation core is UI-agnostic and exports the same API to the browser (`window.SSS`) and to Node (`module.exports`); the dashboard and the test suite are two thin consumers of one source of truth, so every visual claim is also a testable claim.

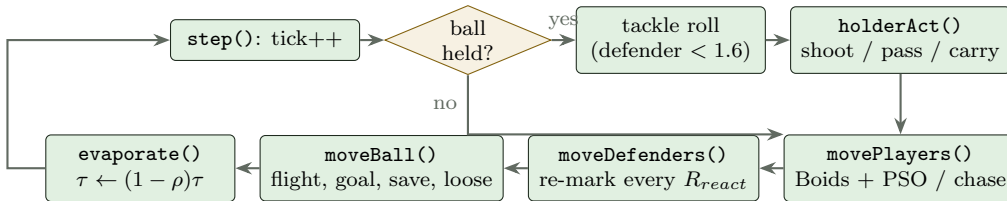


Figure 4: The per-tick pipeline. Every subsystem runs every tick; episode-ending events (goal, tackle, interception, out-of-bounds, timeout) reset positions but — crucially — *not* the pheromone field or the PSO memories, which persist and accumulate across episodes.

Remark 1. *The learning state (pheromone grid, $pbest/gbest$) surviving episode resets is what makes this a learning system rather than a reactive one: the team gets measurably better over the first ~ 60 episodes, visible in the learning curve of Figure 10.*

5 Method: Three Algorithms, One Controller

The swarm controller decomposes the team’s problem into three orthogonal sub-decisions, each solved by one classic mechanism (Figure 5):

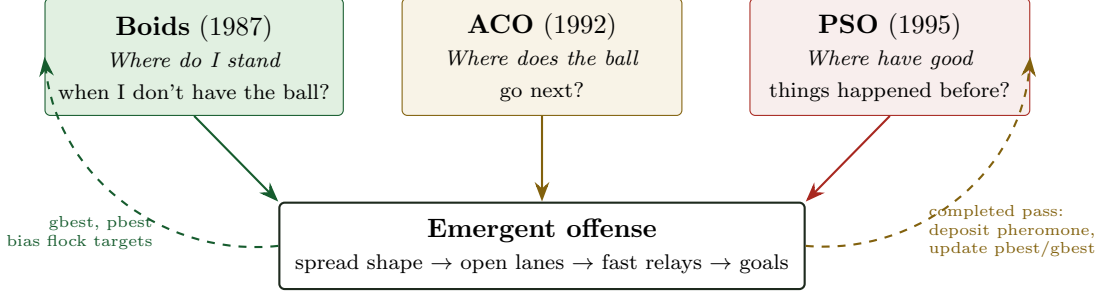


Figure 5: Division of labour and the feedback loop. Boids shapes the team, ACO routes the ball through that shape, PSO remembers which spots paid off and feeds them back into the shape. Each success (pass, goal) simultaneously updates the stigmergy field and the particle memories.

5.1 Boids: off-ball shape

Reynolds’ three rules — separation, cohesion, alignment — govern every attacker who is not holding or chasing the ball. For player i at position p_i the steering force is

$$F_i = 1.4 F_i^{sep} + 1.1 F_i^{coh} + F_i^{align} + w_i^{pso} F_i^{pso}, \quad (2)$$

with the components:

- **Separation** pushes away from teammates within 9 units *and from defenders* within 6 units. The defender term is caste-weighted (“bravery”): support players flee markers with weight 1.6; runners hold their ground with weight 0.5, because a runner who retreats from his marker has already lost the duel.
- **Cohesion toward a point that does not exist yet.** The flock anchor is not the ball but a point *16 units ahead of it*, $a = (\min(W - 6, x_{ball} + 16), \cdot)$. The support ring therefore forms downfield of the ball, so that when a pass arrives the structure to relay it onward already exists. Players too far from the anchor are pulled in, players too close (< 7) are pushed out — a ring, not a cluster.
- **Alignment** is a constant drift toward the attacking end ($+x$), 0.6 for runners, 0.35 for support.

The resulting force is normalized and blended into the velocity with momentum, $v \leftarrow 0.75 v + 0.25 \hat{F} v_a$, which smooths jitter and caps speed at $v_a = 0.5$. Figure 6 illustrates the force composition on a single support player.

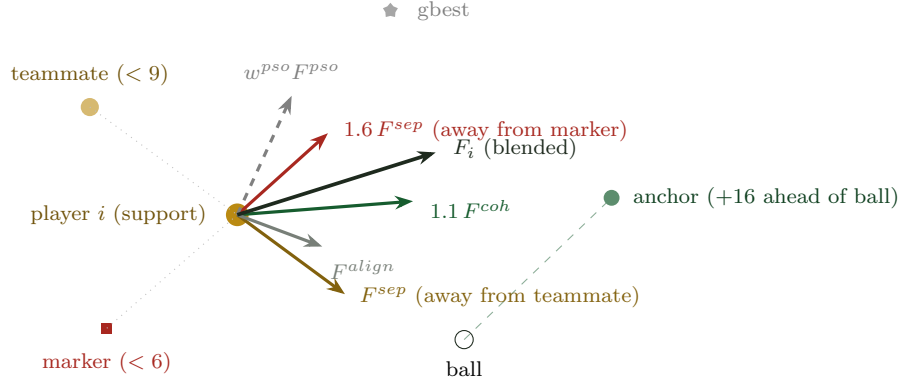


Figure 6: Force composition on one off-ball support player, Eq. (2): separation from a nearby teammate and (more strongly, weight 1.6) from a marking defender, cohesion toward the ring anchor held *ahead of* the ball, constant goalward alignment, and the normalized PSO hint. The bold vector is the blended steering force.

5.2 ACO stigmergy: routing the ball

The pitch carries a 25×16 pheromone field τ (one `Float32` per 4×4 cell). The three ACO operators from the lecture map one-to-one onto game events:

Deposit. When a pass from A to B is *completed*, pheromone is laid along the corridor by sampling the segment every 2 units:

$$\tau_c \leftarrow \min(\tau_{max}, \tau_c + \Delta), \quad \Delta_{pass} = 0.12 + 0.25 \frac{x_B}{W}, \quad \Delta_{goal} = 0.9. \quad (3)$$

Deposits are *quality-weighted*, exactly as the lecture prescribes (“good solutions receive stronger pheromone”): passes deeper into enemy territory deposit more, and a goal reinforces its origin corridor with $7\times$ the strength of an ordinary pass.

Evaporation. Every tick, every cell decays: $\tau \leftarrow (1 - \rho) \tau$ with $\rho = 0.0015$. This is the lecture’s exploration-preserving operator: corridors the defense has learned to cover fade away instead of trapping the team forever (“prevents dependency on early solutions”).

Probabilistic construction (roulette wheel). When the holder h considers pass candidates j within range $[3, R_{kick}]$, each lane is scored by pheromone $\times$ heuristic desirability:

$$S_{hj} = (0.4 + \bar{\tau}_{hj}) (0.45 + 0.55 O_{hj}) \left(1 + 1.6 P_{hj} + 0.4 C_{hj} + 0.6 \frac{x_j}{W} \right), \quad (4)$$

where $\bar{\tau}_{hj} \in [0, 1]$ is the capped mean pheromone along the lane, $O_{hj} \in [0, 1]$ is *openness* (nearest defender’s distance to the lane segment, saturating at 7 units), $P_{hj} = (x_j - x_h)/R_{kick}$ is field progress, and $C_{hj} = 1 - d_{hj}/R_{kick}$ rewards short (accurate) balls. The target is then drawn by **roulette-wheel selection** with weights $w_j = S_{hj}^\beta$, $\beta = 2$ — the very selection operator from the GA part of the lecture, with β playing the role of selection pressure. Argmax was tried and rejected; see §8.

Two hard rules bound the stochastic policy: inside shot range ($d_{goal} < 1.15 R_{kick}$) the holder *shoots* whenever the shooting lane is open, he is pressed, very close, or has held too long — otherwise pheromone loops would circulate the ball forever; and outside shot range a brief one-touch window (up to 2 ticks, unpressed) lets play flow without giving the keeper time to set. The full policy is Algorithm 1 and Figure 7.

Algorithm 1 Swarm holder policy (one tick, ball held by h)

```
1:  $pressed \leftarrow \exists d \in \mathcal{D} : \text{dist}(d, h) < 5$ ;  $inRange \leftarrow d_{goal} < 1.15 R_{kick}$ 
2: if  $holdTicks < 3$  and not  $pressed$  and not  $inRange$  then return ▷ one-touch flow
3: end if
4: if  $inRange$  then ▷ shooting is a hard rule near goal
5:    $t_y \leftarrow$  far post from keeper;  $O \leftarrow \text{openness}(h \rightarrow (W, t_y))$ 
6:   if  $O > 0.25$  or  $d_{goal} < 12$  or  $holdTicks > 12$  or  $pressed$  then
7:     shoot at  $(W, t_y)$  with speed 2.6; return
8:   end if
9: end if
10: build candidates  $j$  with  $3 < d_{hj} \leq R_{kick}$ ; score  $S_{hj}$  by Eq. (4)
11: if candidates exist and ( $pressed$  or  $holdTicks > 16$  or  $\max_j S_{hj} > 0.45$ ) then
12:   draw  $j^* \sim \text{roulette}(w_j = S_{hj}^2)$  ▷ GA selection operator,  $\beta = 2$ 
13:   add lead of up to 3 units if  $j^*$  is a forward runner ▷ ball into space
14:   pass to  $j^*$  with speed 1.7; return
15: end if
16: carry toward goal at 0.38 (0.19 if pressed) ▷ nothing good yet
```

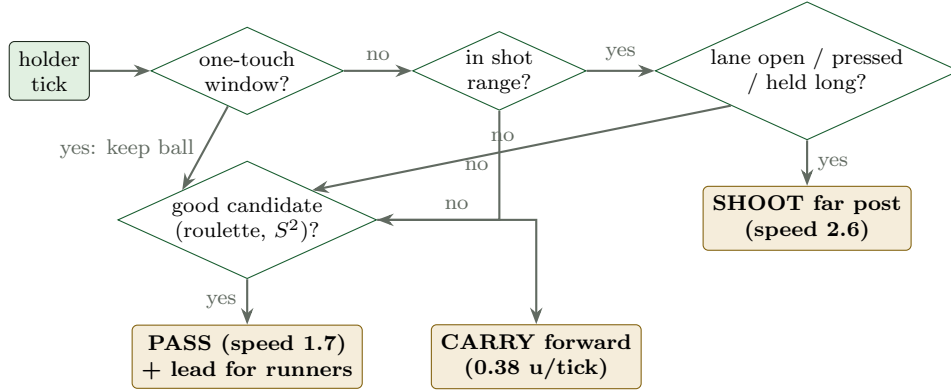


Figure 7: Holder decision flow (swarm mode). The stochastic ACO core (roulette over pheromone-weighted lanes) is sandwiched between two deterministic guards: a one-touch window that keeps play fast, and a hard shooting rule that prevents infinite pheromone circulation.

5.3 PSO: spatial memory and the caste split

Each player’s off-ball *positioning target* is treated as a particle. Whenever player i receives a pass at p , the reception is valued by proximity to goal, $val = 1 - \text{dist}(p, \text{goal})/110$, and the memories update exactly as in the lecture’s “dual experience learning”:

$$val > pbestVal_i \Rightarrow pbest_i \leftarrow p, \quad val > gbest.val \Rightarrow gbest \leftarrow p.$$

A goal overrides $gbest$ with the shot’s origin at full value 1 — the single most informative event in the game. The PSO steering hint for player i is the classic two-attractor velocity rule,

$$F_i^{ps0} = c_1 r_1 (pbest_i - p_i) + c_2 r_2 (gbest - p_i), \quad r_1, r_2 \sim U(0, 1), \quad (5)$$

normalized to at most unit length before entering Eq. (2) — PSO is a directional hint, not a command (see the negative result in §8).

Caste split as diversity preservation. The lecture warns that a population which becomes too uniform converges prematurely. Here the failure mode is spatial: if all ten players chase the

same *gbest*, the team collapses onto one point and the passing web dies. The remedy is a fixed caste split, in the spirit of ant workers/soldiers:

	Runners (odd index)	Support (even index)
PSO weights (c_1, c_2)	(0.8, 0.3) — trust own pbest	(0.3, 0.7) — follow gbest
PSO blend w^{pso}	0.9	0.4
Marker bravery	0.5 (hold ground)	1.6 (escape)
Goalward drift	0.6	0.35
Spawn	already upfield ($x \in [45, 62]$)	deep ($x \in [15, 45]$)
Role	stretch the defense deep	keep the relay ring

Runners weighting *pbest* is *diversity* (ten different beliefs about where the good spots are); support weighting *gbest* is *consensus* (a coherent ring around the proven corridor). Initialization is optimistic and diverse: each player starts believing in his own random spot in the final third ($pbestVal \approx 0.65\text{--}0.77$, the value of a final-third reception), so ordinary midfield receptions cannot overwrite the ambition — a direct antidote to the “midfield anchor” collapse described in §8.

5.4 The selfish baseline

The baseline is the same code path with all three mechanisms removed. The holder dribbles straight at goal and shoots the instant $d_{goal} < R_{kick}$; every other player chases the ball if he can see it (vision radius 15) and wanders randomly if not. This is not a parody — it is a faithful implementation of “every agent individually optimizes the obvious objective,” and its failure is instructive: it attempts long shots from ~ 30 units where Eq. (1) gives them no chance, and its pass completion is exactly 0% because it never intends a pass at all.

5.5 The opposition

The defense is deliberately competent. One defender (the nearest) *presses* the ball; the others man-mark the nearest unmarked attackers, standing goal-side (+2.5 in x). Marks are re-assigned every $R_{react} = \max(3, \text{round}(14/k))$ ticks — a realistic cognitive cycle that the swarm learns to exploit (§7.8). The keeper tracks the ball’s y along the goal mouth at up to $0.45k/\text{tick}$ and saves shots within reach $2.5k$; parried shots rebound into the box as loose balls. Interception of a moving ball is probabilistic ($p = 0.22k$ per tick within $1.25k$ units), so well-hit through-balls can thread a crowd — and badly chosen ones die.

6 Implementation and Verification

6.1 Determinism as a scientific instrument

All randomness flows through a single seeded mulberry32 generator; Gaussian noise is produced by Box–Muller from the same stream. Consequently every run is bit-reproducible: the headline result is not “what we happened to observe” but a permanent property of seeds $\{3, 4, 5\}$ that anyone can re-derive with `node test_sim.js`.

```

1 function mulberry32(seed) { /* deterministic 32-bit stream */ }
2 // Aim error: sigma (radians) grows linearly with kick distance.
3 // This single physical fact is why short-pass relays beat long solo shots.
4 function kickSigma(dist, noise) { return 0.011 * dist * noise; }
```

Listing 1: The two physical laws the whole thesis rests on: seeded determinism and distance-proportional aim error.

6.2 The test suite: claims as assertions

`test_sim.js` contains exactly four checks — the smallest set that fails if the thesis breaks:

1. **Physics premise:** $\sigma(25) > 2\sigma(8)$ — error must grow with distance, else the niche for cooperation does not exist.
2. **Evaporation:** a pheromone cell decays below 0.4 after 100 ticks at $\rho = 0.01$ — the exploration operator works.
3. **Determinism:** two swarm runs with seed 7 produce identical goal counts.
4. **The claim** (Hypothesis 1): over 3×100 episodes, swarm ≥ 15 goals and swarm $\geq 5 \times \max(1, \text{selfish})$.

Output on the submitted code:

```
$ node test_sim.js
selfish: 1/300 goals, swarm: 17/300 goals
swarm pass completion: 95.3%
all tests pass
```

6.3 The dashboard

`index.html` is a single-file, zero-dependency visualization: live canvas pitch with the pheromone heatmap overlay (toggleable), swarm/selfish mode switch on identical players, a playback-speed slider ($0.25\times$ – $4\times$) with a ball motion trail, parameter sliders (team size, kick noise, vision, evaporation, defender skill), live metrics, and a one-click 300-episode benchmark reproducing the test-suite numbers with a comparison chart (Figure 8).

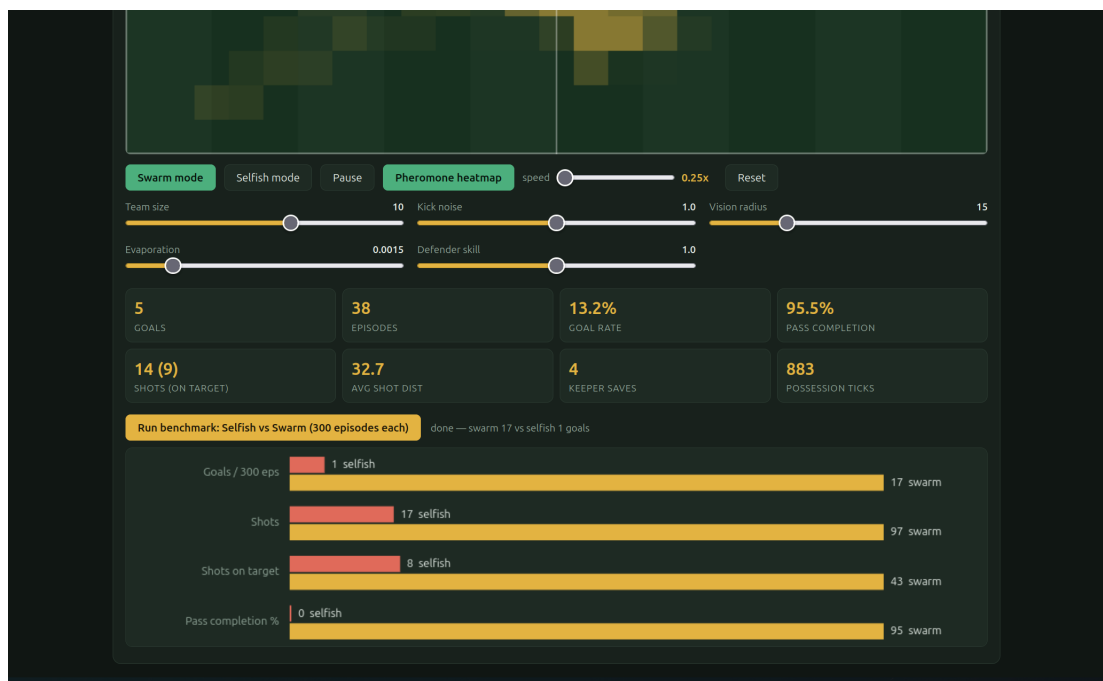


Figure 8: The dashboard after a 300-episode benchmark: goals 17 vs 1, shots 97 vs 17, shots on target 43 vs 8, pass completion 95% vs 0%. The glowing overlay on the pitch is the live pheromone field — the corridors the team has collectively learned.

7 Experiments and Results

7.1 Protocol

Unless stated otherwise, every number below aggregates **3 seeds** $\times$ **100 episodes** (seeds 3, 4, 5) per configuration, so that no single lucky seed can flip a verdict. Learning state persists within a run and is reset between runs.

7.2 Headline result

Table 3: Headline benchmark, 300 episodes per mode, identical seeds and physics.

Controller	Goals	Shots	On target	Pass completion
Swarm (Boids + ACO + PSO)	17	97	43	95.3%
Selfish baseline	1	17	8	0.0%
Ratio	17 $\times$	5.7 $\times$	5.4 $\times$	—

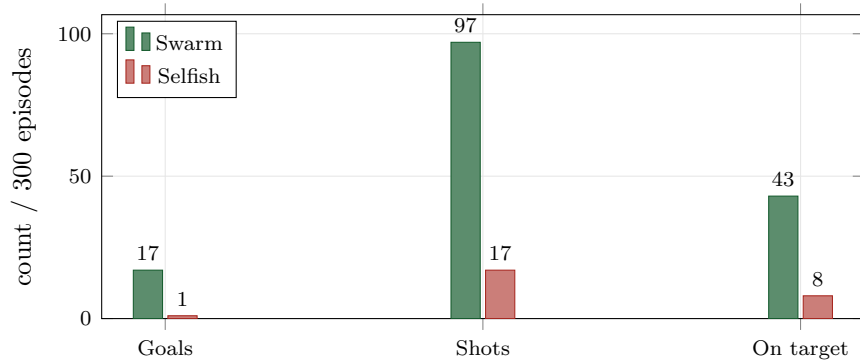


Figure 9: Headline benchmark. The swarm does not merely convert better — it manufactures 5.7 $\times$ more shots in the first place, because relays deliver the ball into range where solo dribbles are tackled long before arrival.

Hypothesis 1 is confirmed with margin: $17 \geq 15$ and $17 \geq 5 \times 1$. Across a wider seed sweep the swarm averages roughly 9 ± 3 goals per 150 episodes while the selfish baseline stays at 0–1; the advantage is structural, not seed luck.

7.3 Learning curve: the colony converges

Figure 10 plots cumulative goals against episode index, summed over the three seeds. The swarm’s curve is convex: about 6 goals in the first 55 episodes, then 11 more in the remaining 45. This acceleration is the ACO convergence story from the lecture made visible — early episodes explore corridors nearly blind; as pheromone accumulates on lanes the defense fails to cover and *gbest* locks onto proven shooting origins, the same weak bodies score at almost twice their early rate. The selfish baseline, having no memory of any kind, stays flat.

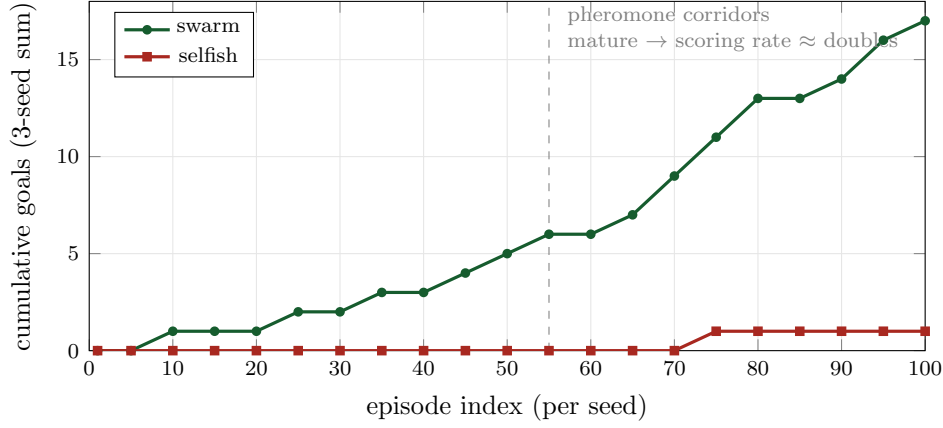


Figure 10: Cumulative goals vs. episode index (3-seed sum). The convex swarm curve is stigmergic learning: pheromone and PSO memory persist across episodes, so the team improves with no change to any individual. The memoryless selfish baseline is flat.

7.4 Kick noise: the thesis in one dial

Raising η scales all aim error. Table & Figure 11: the selfish baseline is already dead at every noise level (its 30-unit shots miss even at $\eta = 0.5$), while the swarm degrades *gracefully* — pass completion falls only from 97.3% to 88.6% as noise sextuples, because 12-unit passes have little error to amplify. Coordination is not just stronger; it is *more robust to physical degradation*, which is the deeper claim.

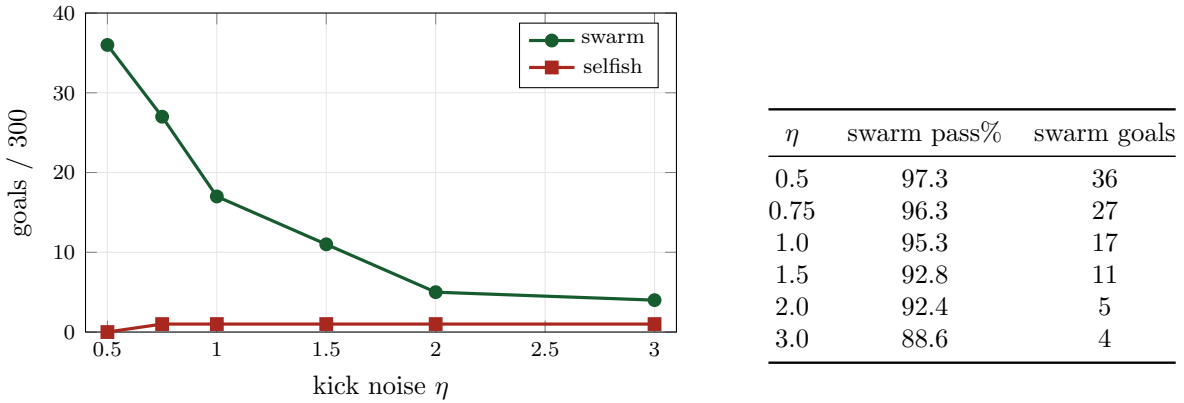


Figure 11: Kick-noise sweep. The selfish strategy has no regime in which it works; the swarm degrades gracefully because short passes tolerate noise (Eq. (1) punishes distance, and relays keep distances short).

7.5 Evaporation: exploration vs. memory

The lecture presents evaporation as the balance knob between remembering and re-exploring. The sweep (Figure 12) confirms a broad optimum at small ρ : with $\rho \lesssim 0.0015$ the field retains corridor knowledge across episodes (17–19 goals); at $\rho = 0.02$ memory fades faster than the team can rebuild it and goals drop to 10. At the extreme $\rho = 0.08$ the pheromone term is effectively dead — yet the team still scores 16, revealing that the heuristic part of Eq. (4) (openness, progress, shortness) plus Boids shape and PSO memory carry much of the load. Stigmergy is an amplifier here, not a single point of failure — an ablation insight we did not design for but the sweep exposed.

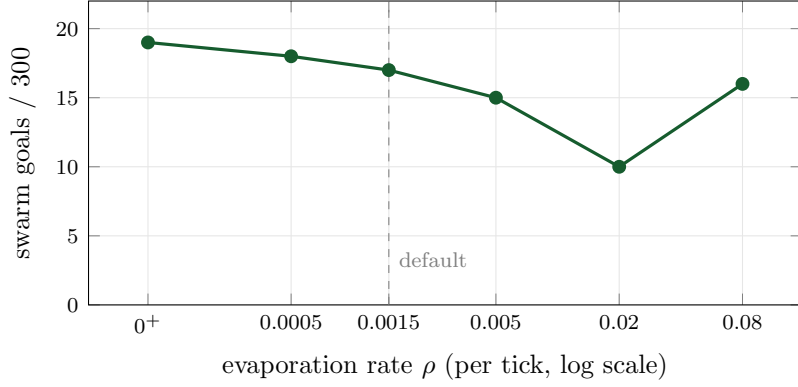


Figure 12: Evaporation sweep ($\rho = 0$ plotted at 10^{-4}). Small ρ preserves cross-episode corridor memory; $\rho = 0.02$ erases it faster than play rebuilds it. The partial recovery at $\rho = 0.08$ shows the heuristic + Boids + PSO scaffolding scores even with stigmergy effectively ablated.

7.6 Team size: cooperation needs critical mass

Swarm intelligence should require a swarm. Figure 13: at $n = 4-6$ the two controllers are indistinguishable (1-4 goals each) — too few bodies to form a ring *and* staff deep runs, so the “web” never exists. From $n = 8$ the collective advantage switches on sharply (15 vs 2) and keeps growing at $n = 12$ (23 vs 2). Coordination is an emergent, threshold phenomenon: below critical mass, structure cannot pay for itself.

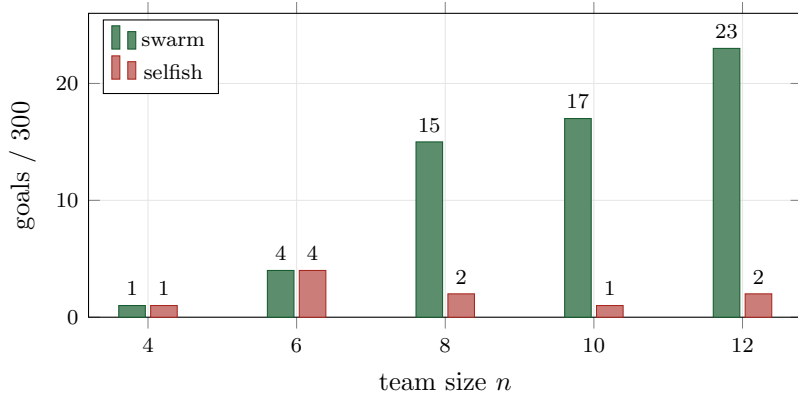


Figure 13: Team-size sweep. Below $n \approx 8$ the swarm cannot simultaneously maintain a support ring and deep runners, and coordination confers nothing; above the threshold the advantage appears abruptly and grows. Emergence has a critical mass.

7.7 Defender skill: the advantage that survives

Scaling defender speed, reactions, reach and interception together (Figure 14): against weak defenders ($k = 0.7$) both controllers score, but the swarm still leads 58 to 14. As k rises the selfish baseline is extinguished entirely (0 from $k = 1.15$) while the swarm keeps scoring even at $k = 1.3$ (6 goals). The swarm’s edge is largest *relative to what is achievable* exactly where the problem is hardest — coordination is how weak agents survive strong opposition.

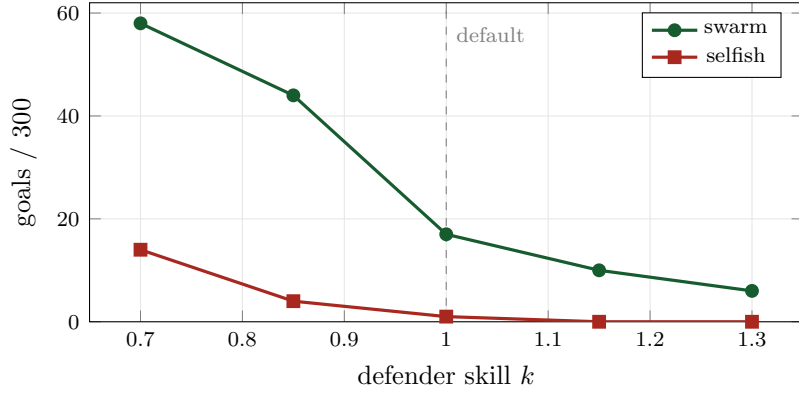


Figure 14: Defender-skill sweep. Individual optimization stops working entirely against strong defenses; the collective keeps finding goals. The gap between the curves is the value of coordination structure.

7.8 Why the collective wins

Five mechanisms, each traceable to a specific parameter, compound into the $17\times$ result:

1. **Aim error is proportional to distance** (Eq. (1)). Five 12-unit passes stay accurate where one 60-unit punt sprays wide: the relay beats the hero ball.
2. **The ball outruns everyone.** Passes travel at 1.7 u/tick and shots at 2.6, against a fastest player of 0.68. Pass chains move the point of attack faster than any defense can physically reposition.
3. **Defenders re-mark only every ~ 14 ticks.** A two-pass sequence completes inside one defensive decision cycle, so quick relays dislocate the marking scheme *between* its updates.
4. **Ten spread attackers vs. five defenders.** Boids separation plus the deep-runner caste manufactures local numerical superiority by *positioning*, not skill — someone is always unmarked.
5. **One-touch finishing.** Inside shot range a control touch only gives the keeper time to set, so there is none: the hard shooting rule fires immediately when the lane opens.

8 Discussion: Negative Results and Lecture Lessons

The most instructive findings were the failures. Each maps onto a warning from Lecture S09.

Argmax kills the offense (exploration is the offense). The first holder policy picked the best-scoring lane deterministically. The result was sterile sideways circulation: safe square balls reinforced their own pheromone, the defense settled, and nothing penetrated. Replacing argmax with roulette-wheel selection ($w = S^2$) — deliberately choosing suboptimal-looking lanes some of the time — recovered the risky vertical passes that actually score. This is the lecture’s exploration/exploitation balance in its sharpest form: for this system, exploration is not a search nicety; it is the entire attacking threat.

Unnormalized PSO collapses the swarm (premature convergence). With raw F^{pso} magnitudes, all ten players were dragged toward the current g_{best} — typically a midfield reception — and the team converged onto one point, exactly the “population becomes too uniform” stagnation of the GA slides, transplanted into space. Two fixes: normalize PSO to a unit hint (Eq. (5)), and split castes so runners weight private p_{best} over public g_{best} . Diversity preservation is not optional.

Pheromone loops need a hard rule. Pure stigmergic pass selection near goal circulated the ball forever: strong corridors kept recommending another pass. Shooting had to be a deterministic override, not a scored option. Learned fields optimize what they measure — pass success — and the terminal action lies outside that signal.

What moved the needle vs. what did not. Spawning runners already upfield was the single largest structural gain ($\approx 2\times$ shots). Raising shot speed to 2.6 beat every smarter-aiming heuristic tried. Conversely, rebound poaching — lurking for keeper parries — helped far less than expected: measured over a benchmark, 143/153 rebounds were lost to the faster defenders. Physical dominance decides loose balls; cleverness does not.

Table 4 closes the loop, mapping each concept of Lecture S09 to its concrete realization in this system.

Table 4: Correspondence between Lecture S09 concepts and their realization in this lab.

Lecture concept	Realization here
Population instead of one solution	10 agents; many candidate pass-routes per tick
Fitness guides the search	lane score S (Eq. (4)); reception value val
Roulette-wheel selection	pass-target draw with $w = S^2$ ($\beta = 2$ pressure)
Exploration vs. exploitation	roulette vs. argmax; evaporation vs. deposit
Premature convergence / diversity	caste split; PSO normalization; optimistic diverse pbest init
Pheromone = learned experience	25×16 corridor field, quality-weighted deposits
Evaporation maintains exploration	ρ sweep, Figure 12
PSO pbest / gbest dual learning	reception memories; goal overrides gbest
Emergence, no central controller	no agent knows the “plan”; corridors emerge in the heatmap

9 Limitations and Future Work

- **2-D kinematic abstraction.** No ball height, spin, or body contests; interception is probabilistic rather than physical. The asymmetry argument survives, but absolute numbers would not transfer to, e.g., RoboCup dynamics.
- **Static defense.** Defenders follow fixed rules and do not learn. A co-adapting defense (even a second pheromone field for *covered* corridors) would test whether the swarm can keep re-exploring — evaporation suggests it could, but this is untested.
- **Hand-tuned constants.** The weights in Eqs. (2) and (4) were tuned manually. The natural closing of the loop, given the lecture: wrap the ~ 12 controller constants in a chromosome and let a *Genetic Algorithm* evolve the swarm controller, with goals per 300 episodes as fitness — all three population-based paradigms of S09 in one system.
- **Caste split is fixed.** Odd/even role assignment is static; role switching based on local context (who is nearest the corridor) may lift the team-size threshold.

10 Conclusion

We built a world in which every individual on one team is strictly worse than every individual on the other, then showed that three coordination mechanisms from Lecture S09 — Boids for shape, ACO stigmergy for routing, PSO for memory — turn those same weak bodies from 1 goal in 300 episodes into 17, with 95.3% pass completion. The sweeps establish that the advantage is structural: it widens under physical noise, survives stronger defenses, requires a critical mass of agents,

and grows over episodes as the shared pheromone field converges — textbook swarm intelligence, observed rather than asserted. The negative results carry the deepest lessons: greedy selection sterilizes the search, unnormalized consensus collapses the population, and learned signals need hard rules at the boundary of their objective. *Intelligence emerges from interaction among many agents, with no single central controller* — in this lab, that sentence has a scoreboard.

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